How Fast Does a Monopile Wobble?

Natural frequency of a realistic offshore wind foundation at each installation stage

Project	Vibration and fatigue estimation of a bare offshore monopile before commissioning (self-study)
This phase	Extend the validated beam model to a realistic monopile with soil along the pile, and compare four installation stages
Tools	Python 3 (NumPy, SciPy, Matplotlib); script: phase_a_monopile.py
Status	Phase A complete. All geometry, mass and soil values are illustrative, not site data.

1. The idea in simple words

Picture the turbine as a **tall ruler clamped in a table**. If you flick it, it wobbles back and forth at a certain speed. That speed is the **natural frequency**, measured in wobbles per second (Hz). It matters because if waves or wind push the structure at about this speed, the motion gets amplified and fatigue damage grows.

In Phase A-1 the 'ruler' was a simple 60 m tube. Here it is a realistic foundation: a pile in the seabed, a transition piece, a tapered tower and a turbine on top. The question: **how does the wobble speed change as each part is added?**

2. Key findings at a glance

#	Finding	In simple words
1	Every installed part lowers the natural frequency	1.22 Hz (bare pile), 0.86 (+ transition piece), 0.51 (+ tower), 0.26 Hz (full turbine). More height and more mass mean a slower wobble.
2	Soft soil and the water around the pile both lower it	Frequency depends on stiffness and mass: stiffer is faster, heavier is slower. Soft soil makes the base less stiff, because it can rock in the ground. Water around the pile makes it effectively heavier, because the water has to move with it. Both effects are large for the bare pile (soft vs. stiff soil: 87 %, water: 36 %) and small for the finished turbine (21 % and 0.4 %), where tower and rotor decide everything.
3	Mode 1 is the important bending shape	Higher modes are simply wavier shapes with more crossing points and much higher frequencies (for the bare pile, mode 2 is 7.6 Hz), far from the wave band. The results contain no special frequency limit such as 3 Hz.
4	No frequency limit can be chosen; each stage must be checked	Each stage has the frequency its parts give it. What can be done is to compare every stage that stays standing with what excites it: waves (0.08 to 0.17 Hz) and wind vortices (critical wind speed U = f x D / 0.2). The tower without the rotor (about 9 to 19 m/s) is the stage to watch; it should be kept short or protected (for example with a damper or strakes).

Phase A has no loads and no fatigue yet, so finding 4 is a hypothesis that Phases B and C will test.

3. What was modelled

Part	What it is	Key values (illustrative)
Pile	Steel tube, 30 m in the seabed, 30 m in water, 10 m above water	D = 8.0 m, t = 90 mm
Transition piece	Connection between pile and tower, up to 55 m	D = 8.4 m, t = 80 mm, plus 200 t platform mass
Tower	Tapered tube from 55 m to hub at 140 m	D = 6.5 to 4.5 m, t = 45 to 22 mm
Nacelle and rotor	Lumped mass at the top	450 t
Soil	Springs along the buried pile, stiffer with depth ($k = n_h$ x depth)	Soft 5, medium 15, stiff 40 MN/m^3
Water	Water moving with the pile ('added mass') on the submerged part	Ca = 1.0

Steel mass of the structure is about 1870 t (plus the platform and rotor-nacelle mass), a sensible size for an 8 MW class turbine. The model is the same beam finite element code as in Phase A-1, now with a different tube size per element, soil springs along the pile, and lumped masses.

4. Check: is the code still correct?

The new code was first run on the old test case (60 m tube, fixed base). It gives 1.686 / 10.564 / 29.580 Hz, identical to the analytical result. So the extension did not break anything, and differences below come from the physical inputs, not from coding errors.

5. Result 1: every added part slows the wobble

Installation stage	First frequency (Hz)	Change vs. previous stage	Wobble period
Bare monopile	1.22	-	0.8 s
+ Transition piece	0.86	-29 %	1.2 s
+ Tower	0.51	-41 %	2.0 s
+ Nacelle and rotor (full turbine)	0.26	-50 %	3.9 s

Medium soil, with added mass.

Like a ruler that gets longer and gets weights on top, the structure swings more slowly at each stage. The full turbine swings about once every 4 seconds, at the lower edge of the 0.25 to 0.35 Hz range typical for real turbines. This is a good sign that the model is realistic.

> **Main message:** the same foundation behaves very differently at each stage. The bare pile is about five times faster than the finished turbine.

6. Result 2: how the structure bends (mode shapes)

A mode shape shows *how* the structure bends when it wobbles at one of its natural frequencies. Each plot is one stage; the vertical axis is height, the horizontal axis is how far each height moves (scaled so the largest move is 1). Mode 1 (blue) is the simple 'lean one way, then the other' motion and is the most important one.

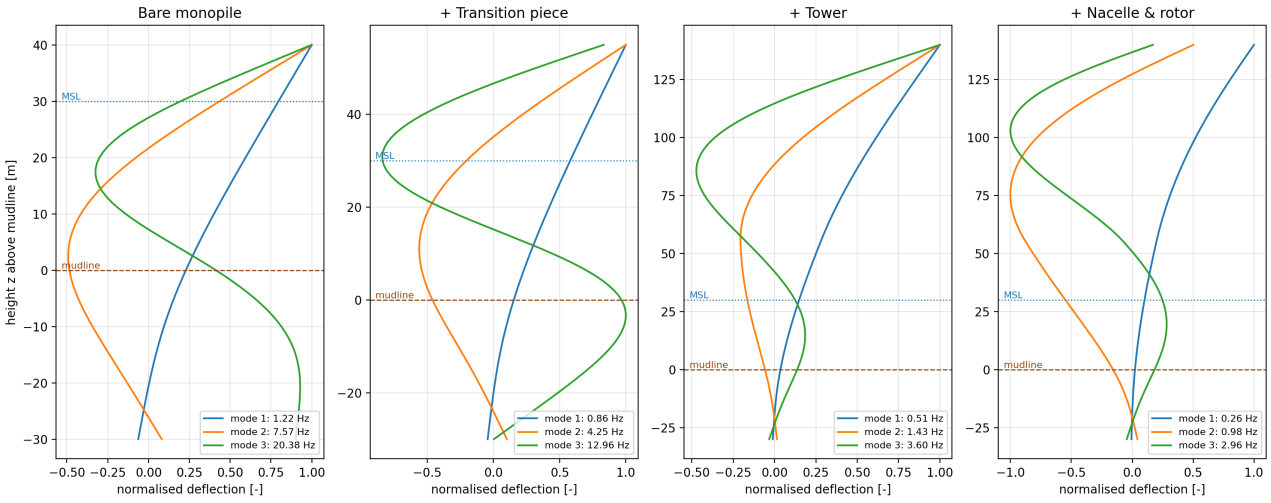


Figure 1: First three mode shapes for each installation stage (medium soil).

- Mode 1 is a smooth lean of the whole structure. The top moves most, the pile in the seabed hardly moves.
- Higher modes (orange, green) have one or more crossing points. They are much faster: for the full turbine mode 2 is about 1 Hz, about four times mode 1.
- For the bare pile, modes 2 and 3 are above 7 Hz, far from anything waves excite, so one mode is a good approximation. For the tall stages mode 2 lies closer and should be mentioned as a limitation.

7. Result 3: soft soil also slows the wobble

Think of the ruler clamped in a sponge instead of a table. A softer sponge lets the base rock, so the wobble slows down.

Stage	Soft soil (Hz)	Medium (Hz)	Stiff soil (Hz)	Stiff / soft	Rigid base (Hz)
Bare monopile	0.83	1.22	1.54	1.9 x	3.66
+ Transition piece	0.61	0.86	1.04	1.7 x	1.89
+ Tower	0.40	0.51	0.57	1.4 x	0.73
+ Nacelle and rotor	0.22	0.26	0.27	1.2 x	0.30

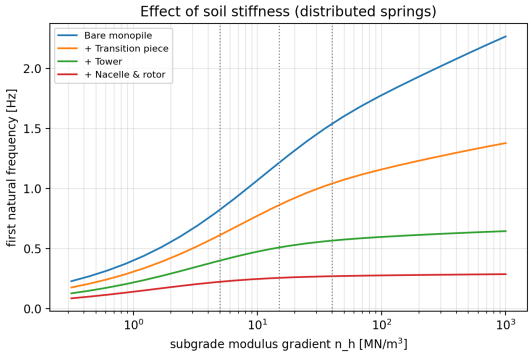


Figure 2: First frequency versus soil stiffness. Dotted lines mark the soft, medium and stiff cases.

- **Soil matters most for the short, bare pile.** Going from soft to stiff soil changes its frequency by 87 %, but only 21 % for the full turbine.
- For the full turbine, assuming a rigid base (0.30 Hz) is only 15 % off, because the slender tower bends much more than the soil yields.
- **Consequence for later phases:** uncertainty in soil stiffness matters most during early installation, which is exactly the period this project studies. It must be included in the Monte Carlo study.

8. Result 4: water around the pile makes it heavier

Water around the submerged pile has to move with it, like swimming in a coat. For the 8 m pile this adds about 51 t per metre, roughly three times the steel mass per metre (about 18 t/m).

Stage	Without water (Hz)	With water (Hz)	Change
Bare monopile	1.91	1.22	-36 %
+ Transition piece	1.02	0.86	-15 %
+ Tower	0.53	0.51	-4 %
+ Nacelle and rotor	0.26	0.26	-0.4 %

Added mass is essential for the bare pile (ignoring it would overestimate the frequency by about 57 %) and almost irrelevant for the finished turbine, where tower and rotor dominate the mass. Phase A-1 ignored it, so this is a clear improvement.

9. Is a low or a high natural frequency better?

Neither is better in general. What matters is how far the frequency is from the frequencies that push the structure. Think of pushing a child on a swing: pushing at the swing's own rhythm makes it go high (resonance); pushing much slower or much faster does little.

	Higher frequency (stiffer)	Lower frequency (more flexible)
Waves (0.08 to 0.17 Hz)	Far above the wave band: waves act like a slow, almost steady push with little amplification	Closer to the wave band: waves can amplify the motion and add fatigue damage

	Higher frequency (stiffer)	Lower frequency (more flexible)
Wind vortices (VIV)	Critical wind speed is high (bare pile about 33 to 62 m/s), which almost never occurs	Critical wind speed falls into normal winds (tower without rotor about 9 to 19 m/s)
Cost	More steel is needed	Less steel is needed

For this project: the bare pile (about 1.2 Hz) is on the safe side for waves and VIV. The tower without the rotor (about 0.4 to 0.57 Hz) is the weak case, because its frequency is low enough for VIV. The finished turbine (0.26 Hz) is a compromise that real designs accept; it must also stay clear of the rotor's own turning frequency, which Phase A does not cover.

> **Rule of thumb:** the best design is not 'as low' or 'as high' as possible, but one whose frequency stays clear of the wave and wind excitation. Phases B and C will show which stage collects the most fatigue damage.

10. What this means for Phase B (waves and vibration)

Waves. Typical North Sea wave energy lies at about 0.08 to 0.17 Hz (wave periods of 6 to 12 s).

- Bare pile and pile with transition piece (about 0.6 to 1.5 Hz): well above the wave band. The structure follows the waves almost statically, with little amplification.
- Full turbine (0.22 to 0.27 Hz): only just above the wave band, so some amplification is possible in short-period seas.

Wind vibration (VIV). Wind flowing around a tube sheds vortices at a rate $f_s = 0.2 \times U / D$. Trouble starts when f_s matches a natural frequency, i.e. at the critical wind speed $U = f \times D / 0.2$ (hand estimate, to be checked in Phase B):

Stage	f_1 range (Hz)	Diameter (m)	Critical wind speed	Verdict
Bare pile	0.8 to 1.5	8.0	about 33 to 62 m/s	Too high, no concern
+ Transition piece	0.6 to 1.0	8.4	about 26 to 44 m/s	Too high, no concern
Tower without rotor	0.40 to 0.57	4.5 to 6.5	about 9 to 19 m/s	**Within normal wind: check closely**

> **Take-away for Phase B:** the stage to examine most carefully is the **tower standing without the rotor-nacelle assembly**. The Strouhal number (0.2) must be verified against a standard or textbook before reporting.

11. Limitations (to state in the final report)

- Soil is linear (springs growing with depth). Real soil is nonlinear and weakens under cyclic loading.
- Transition piece is simply stacked on the pile (no grouted overlap). Tower and platform masses are generic.
- Euler-Bernoulli beams slightly overestimate the stiffness of the stocky buried pile (length/diameter about 4); Timoshenko elements would be better.
- The rotor is a point mass without rotor inertia, and there is no damping and no wind loading yet.
- All values are illustrative and not from a real site, so frequencies are for understanding, not for design. Quote them to two decimals only.

12. Next steps

- Phase B: JONSWAP wave spectrum, Morison wave force and a proper VIV check, using f_1 from the table above.
- Phase C: response, stress at the mudline weld, rainflow counting and fatigue damage for a 6-month waiting period.
- Phase D: logger specification and Monte Carlo study on soil stiffness, damping and wave height.